

Using Local Dynamics to Explain Analog Forecasting of Chaotic Systems

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(Manuscript received 13 July 2020, in final form 11 March 2021)

ABSTRACT: Analogs are nearest neighbors of the state of a system. By using analogs and their successors in time, one is able to produce empirical forecasts. Several analog forecasting methods have been used in atmospheric applications and tested on well-known dynamical systems. Such methods are often used without reference to theoretical connections with dynamical systems. Yet, analog forecasting can be related to the dynamical equations of the system of interest. This study investigates the properties of different analog forecasting strategies by taking local approximations of the system's dynamics. We find that analog forecasting performances are highly linked to the local Jacobian matrix of the flow map, and that analog forecasting combined with linear regression allows us to capture projections of this Jacobian matrix. Additionally, the proposed methodology allows us to efficiently estimate analog forecasting errors, an important component in many applications. Carrying out this analysis also makes it possible to compare different analog forecasting operators, helping us to choose which operator is best suited depending on the situation. These results are derived analytically and tested numerically on two simple chaotic dynamical systems. The impact of observational noise and of the number of analogs is evaluated theoretically and numerically.

KEYWORDS: Statistical techniques; Statistical forecasting; Data science; Machine learning

1. Introduction

To evaluate the future state of a physical system, one strategy is to use physical knowledge to build differential equations that emulate the dynamics of this system. Then, measurements provide information on the initial state from which these equations must be integrated. Data assimilation (Carrassi et al. 2018) gives a framework to account for two main types of error in this forecasting process. First, the aforementioned equations do not describe perfectly the real dynamics of the system, and solving these equations often requires additional approximations, such as numerical discretization. These first error sources combine into what is called model error. Second, observations are usually partial and noisy, such that the initial state from which the differential equations must be integrated is uncertain. Observation error is especially important for chaotic dynamical systems as the latter are highly sensitive to initial conditions. The quantification of model and observational uncertainties is an important topic in data assimilation (Tandeo et al. 2020).

For complex, highly nonlinear systems such as the atmosphere, forecasts based on physical equations are challenging. Indeed, the numerical integration of discretized physical equations can be expensive for high-resolution grids, and unresolved small-scale processes such as convection (Prein et al. 2015) or orographic drag (Lott and Miller 1997) must be parameterized. Therefore, many empirical methods have been used in atmospheric sciences (see Van den Dool 2007

and references therein). The volume of data from model outputs, observations, or the combination of both has increased exponentially in the last decades (Balaji 2015), which has strengthened the scientific interest for empirical methods. One of such methods is called analog forecasting and is based on a notion originally introduced by Lorenz (1969) to estimate atmospheric predictability. Analog forecasting has been used in meteorological applications and on low-dimensional dynamical systems. Analogs were originally used for weather forecasting (Schuurmans 1973), but nowadays numerical weather prediction is usually based on physical models combined with observations through data assimilation, thanks to the progresses of physical models and computer power. However, analog forecasting remains interesting to perform statistical tasks because it is computationally cheap, or when no physical model is available. For instance, Yiou (2014) uses analogs in the context of stochastic weather generators. Several authors (Tandeo et al. 2015; Hamilton et al. 2016; Lguensat et al. 2017; Grooms 2021) combined analog forecasting and data assimilation. Analog forecasting procedures have recently been used in a large range of environmental applications, from well-known atmospheric oscillations (Alexander et al. 2017; Wang et al. 2020) to surface wind velocity (Delle Monache et al. 2013) or solar irradiance (Ayet and Tandeo 2018).

Analog forecasting proposes to bypass physical equations and to use existing trajectories of the system instead, drawing either from numerical model output, observation data or reanalysis. Analog methods are based on the hypothesis that one is provided with many (or one long) trajectories of the

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system of interest, which enables to find analog states close to any initial state, and to use the time successors of these analogs to evaluate the future state of the system. The fluctuating quality and density of available trajectories adds variability to this process. This leads to analog forecasting errors, which play the same role as the previously described model errors. The present study focuses on analog forecasting errors.

Data-based strategies have been used extensively for the forecast of dynamical systems, some of which are similar to the analog methods studied here. Early approaches similar to the locally linear analog forecasting operator (introduced later on in the present paper) include the reconstruction of equations of motion from [Crutchfield and McNamara \(1987\)](#) and the forecast strategy of [Farmer and Sidorowich \(1987\)](#) and [Farmer and Sidorowich \(1988\)](#). In particular, the latter are able to estimate the scaling of forecast errors with forecast time, number of data, and intrinsic properties of the system such as the maximal Lyapunov exponent. This approach is similar to the approach presented in this paper, although we also focus on forecasts at small time horizon, for which the divergence associated with the maximal Lyapunov exponent is not dominant yet. Local approaches have also been combined with time embeddings for data-based forecasts of dynamical systems, a strategy that has grown popular since the early works of [Sauer \(1994\)](#) and [Sugihara \(1994\)](#).

Theoretical arguments on the convergence of analog forecasts toward the real future state can be found in [Zhao and Giannakis \(2016\)](#), where the authors focus on specific kernels that make it possible to enhance analog forecasting performances. More recently, [Alexander and Giannakis \(2020\)](#) showed theoretical convergence toward optimal forecasts for an extended version of conventional analog forecasting approaches, involving kernel principal component regressions. Our objective here is to relate the errors of the analog forecasting operators of [Lguensat et al. \(2017\)](#) to local dynamical properties of the system. In the following, “analog forecasting” refers to the specific type of methods studied by [Lguensat et al. \(2017\)](#). Our work has strong connection with the work of [Atencia and Zawadzki \(2017\)](#), which focuses on the growth of analog forecasting errors to compare analog ensembles with other ensemble forecast methods.

Preliminary results suggest that analog forecasting errors can be estimated empirically using local approximations of the true dynamics ([Platzer et al. 2019](#)). The current paper gives a more in-depth description of the theory that supports different analog forecasting procedures and allows us to evaluate the evolution of analog covariance matrices. The methodology is applied to two chaotic Lorenz systems.

A theoretical framework for analog forecasting is outlined in [section 2](#), and three analog forecasting operators are recalled. Using this framework, the successor-to-future state distance is expressed in [section 3](#). [Section 4](#) examines analog forecasting mean and covariance and investigates the link between linear regression in analog forecasting and the Jacobian matrix of the real system’s flow map.

2. Analog forecasting

a. Mathematical framework

Let a dynamical system be defined by the following time differential equation:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}), \quad (1)$$

where $\mathbf{x}$ is a vector that fully characterizes the state of the system, and $\mathbf{f}$ is a deterministic, vector-valued map. The space $\mathcal{P}$ in which $\mathbf{x}$ lives is called phase space. For real geophysical systems, the physical space is an infinite-dimensional function space, as variables such as pressure or temperature are continuous functions of space and time. Mathematical studies on the attractors from atmospheric systems have suggested a finite dimension, although upper bounds could not be obtained ([Temam 1988](#)). Therefore, an infinite-dimensional physical space yields a finite dimensional phase space for many examples of hydrodynamical systems. Throughout this study, $\mathcal{P}$ is a vector space of finite dimension n . The system is supposed to be autonomous, such that $\mathbf{f}: \mathcal{P} \rightarrow \mathcal{P}$ does not depend on time.

Given an initial state $\mathbf{x}_0$, a forecast gives an estimation of the state of the system $\mathbf{x}_t$ at a later time t . The true future state $\mathbf{x}_t$ is given by the flow map $\Phi: \mathcal{P} \times \mathbb{R} \rightarrow \mathcal{P}$ such that

$$\Phi': \mathbf{x}_0 \rightarrow \Phi'(\mathbf{x}_0) = \mathbf{x}_t. \quad (2)$$

For the dynamical system defined through Eq. (1), Φ represents the time integration of this equation. The theorem of [Poincaré \(1890\)](#) states that trajectories will come back infinitely close to their initial condition after a sufficiently long time and will do so an infinite number of time. It is valid for many geophysical systems of interest and based on the hypothesis that the system can only access a finite volume of phase space, and that the flow “preserves volumes” in phase space. Furthermore, if the dynamical system has an attractor set $\mathcal{A} \in \mathcal{P}$, then almost all trajectories issued from its “basin of attraction” (a set of typical initial conditions) converge to this subset $\mathcal{A}$ ([Milnor 1985](#)). In the following, we assume the simple case that the system has one attractor $\mathcal{A}$, and that all trajectories are in $\mathcal{A}$.

Analog methods are based on the idea that if one is provided with a long enough trajectory of the system of interest, one will find analog states close to any initial point $\mathbf{x}_0$ in the attractor $\mathcal{A}$. The trajectory from which the analogs are taken is called the “catalog” $\mathcal{C}$, and can either come from numerical model output or reprocessed observational data.

Analog forecasting thus supposes that we know a finite number of initial states that are close enough to $\mathbf{x}_0$ to be called “analog,” and that the flow map of the analogs resembles Φ . Therefore, the time successors of the analogs should allow us to estimate the real future state $\mathbf{x}_t$. In the following, the k th analog and its successor are noted $\mathbf{a}_0^k$ and $\mathbf{a}_t^k$. Note that analog forecasting is intrinsically random as it depends on the catalog, which is one out of many possible trajectories. The variability in the catalog influences the ability of the analogs and successors to estimate the future state. This motivates the use of probabilistic analog forecasting operators Θ such that

$$\Theta^t: \mathbf{x}_0 \rightarrow \Theta^t(\mathbf{x}_0), \quad (3)$$

where $\Theta^t(\mathbf{x}_0)$ is a distribution that gives information both about the estimation of the future state $\mathbf{x}_t$ and the variability of this estimation process.

Note that for chaotic dynamical systems, any small but finite perturbation in the initial condition will grow exponentially in time. Therefore, however small the distance between the analog $\mathbf{a}_0^k$ and the initial state $\mathbf{x}_0$ may be, the distance between the successor $\mathbf{a}_t^k$ and the future state $\mathbf{x}_t$ will eventually be as large as the typical distance between two random points on the attractor. For a chaotic system, we define the Lyapunov time as the time it takes for an infinitesimal error in initial conditions to double. The time needed for $|\mathbf{a}_t^k - \mathbf{x}_t|$ to be as large as the distance between any two points in the attractor is a function of both the Lyapunov time and the initial distance between $\mathbf{a}_0^k$ and $\mathbf{x}_0$. After this divergence, the analog forecast is as informative as a simple climatological forecast (i.e., a forecast based on average statistics on the attractor). This study is devoted to the properties of analog forecasts when the distance between $\mathbf{a}_t^k$ and $\mathbf{x}_t$ is still small compared to the typical distance between any two points on the attractor.

In atmospheric applications, the Lyapunov time is of the order of 5–10 days (see Vannitsem 2017 and references therein). The chaotic behavior makes forecasts difficult, whatever the methodology used. It is the main limiting factor of atmospheric predictability horizon. For a more detailed introduction to Lyapunov exponents and chaos theory in the context of atmospheric applications, the reader is referred to Dijkstra (2016). Note that the quantification of atmospheric predictability was the first motivation for the study of atmospheric analogs (Lorenz 1969), and the latter are still used in this purpose nowadays (Li and Ding 2011).

b. Analog forecasting operators

We present three analog forecasting operators originally introduced in Lguensat et al. (2017). A finite number K of analogs $(\mathbf{a}_0^k)_{k \in [1, K]}$ and successors $(\mathbf{a}_t^k)_{k \in [1, K]}$ are used and are assigned weights $(\omega_k)_{k \in [1, K]}$.

The distributions of each analog forecast $\Theta^t(\mathbf{x}_0)$ is a discrete distribution with finite support (i.e., a finite sum of Dirac delta functions), with each pair of analog/successor defining an element of the empirical distribution.

The locally constant operator (“LC operator” or simply “LC”) uses only the successors to estimate $\mathbf{x}_t$. $\Theta_{LC}^t(\mathbf{x}_0) \sim \sum_k \omega_k \delta_{\mathbf{a}_t^k}(\cdot)$. The mean forecast is thus $\mu_{LC} = \sum_k \omega_k \mathbf{a}_t^k$. The covariance of the forecast is $\text{cov}_{\omega_k}(\mathbf{a}_t^k)$, the ω -weighted empirical covariance of the successors.

The locally incremental operator (“LI operator” or simply “LI”) uses $\mathbf{x}_0$, the analogs and the successors to estimate $\mathbf{x}_t$. $\Theta_{LI}^t(\mathbf{x}_0) \sim \sum_k \omega_k \delta_{\mathbf{x}_0 + \mathbf{a}_t^k - \mathbf{a}_0^k}(\cdot)$. The mean forecast is $\mu_{LI} = \mathbf{x}_0 + \sum_k \omega_k (\mathbf{a}_t^k - \mathbf{a}_0^k)$. The covariance of the forecast is $\text{cov}_{\omega_k}(\mathbf{a}_t^k - \mathbf{a}_0^k)$, the ω -weighted empirical covariance of the increments.

The locally linear operator (“LL operator” or simply “LL”) performs a weighted linear regression between the analogs and the successors. The regression is applied between

$\mathbf{a}_0^k - \mu_0$ and the successors $\mathbf{a}_t^k$, where $\mu_0 = \sum_k \omega_k \mathbf{a}_0^k$. This gives slope $\mathbf{S}$, intercept $\mathbf{c}$, and residuals $\xi^k = \mathbf{a}_t^k - \mathbf{S}(\mathbf{a}_0^k - \mu_0) - \mathbf{c}$. $\Theta_{LL}^t(\mathbf{x}_0) \sim \sum_k \omega_k \delta_{\mu_{LL} + \xi^k}(\cdot)$. The mean forecast is $\mu_{LL} = \mathbf{S}(\mathbf{x}_0 - \mu_0) + \mathbf{c}$. The covariance of the forecast is $\text{cov}_{\omega_k}(\xi^k)$, the ω -weighted empirical covariance of the residuals.

The LC, LI, and LL analog forecasting operators are illustrated in Fig. 1. The variance of the LC is similar around $t = 0$ and for the final value of t . On the other hand, the variance of the LI goes to 0 as $t \rightarrow 0$, but for large times the LI estimator has a larger variance compared to the LC. The next sections provide some information that helps to interpret this phenomenon. The LL is able to catch the dynamics, and therefore shows a smaller variance and better skills than the LC and LI at all times. This is due to the fact that, in this example, nonlinear terms are negligible, and the flow map of the analogs matches exactly the real system’s flow map.

It is worth mentioning another kind of analog forecasting operator called “constructed analogs” (CA). It is a particular case of the LC operator where the weights ω_k^{CA} can have negative values and are such that the mean of the analogs μ_0 is as close as possible to the initial state: $\{\omega_k^{CA}\}_k = \text{argmin}_{\omega_k} |\sum_k \omega_k \mathbf{a}_0^k - \mathbf{x}_0|$. It was used by Van Den Dool (1994) in cases where using directly the analogs from the catalog did not yield satisfactory results. Later, Tippett and DelSole (2013) showed that CA are equivalent to the LL operator with constant weights.

In the following and unless otherwise specified, it is assumed that the weights ω_k are positive and decreasing functions of the distance between $\mathbf{a}_0^k$ and $\mathbf{x}_0$, and that $\sum_k \omega_k = 1$. This is a common choice, for instance, Lguensat et al. (2017) set $\omega_k \propto \exp(-|\mathbf{a}_0^k - \mathbf{x}_0|^2 / 2l^2)$ where l is the median of analog-to-initial state distances. This allows us to give more weight to the analogs that are closest to the real initial state. On the same topic, see the work of McDermott and Wikle (2016). However, the present analysis of analog forecasting errors is not restricted to any particular choice of weights. In the following, we make approximations in the limit $\mathbf{a}_0^k \rightarrow \mathbf{x}_0$, so that the only hypothesis concerning the weights is that $\omega_k \rightarrow 0$ if the k th analog is too far away from $\mathbf{x}_0$ for the limit to be taken.

3. Successor-to-future state distance

a. Notations and hypotheses

This work assumes that the evolution dynamics of the analogs are similar to the evolution dynamics of the system of interest, and that the system is deterministic. This can be stated in a differential equation form:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}) \\ \mathbf{x}_{t=0} = \mathbf{x}_0 \end{cases}, \quad (4a)$$

$$\forall k, \begin{cases} \frac{d\mathbf{a}^k}{dt} = \mathbf{f}_a(\mathbf{a}^k) \\ \mathbf{a}_{t=0}^k = \mathbf{a}_0^k \end{cases}, \quad (4b)$$

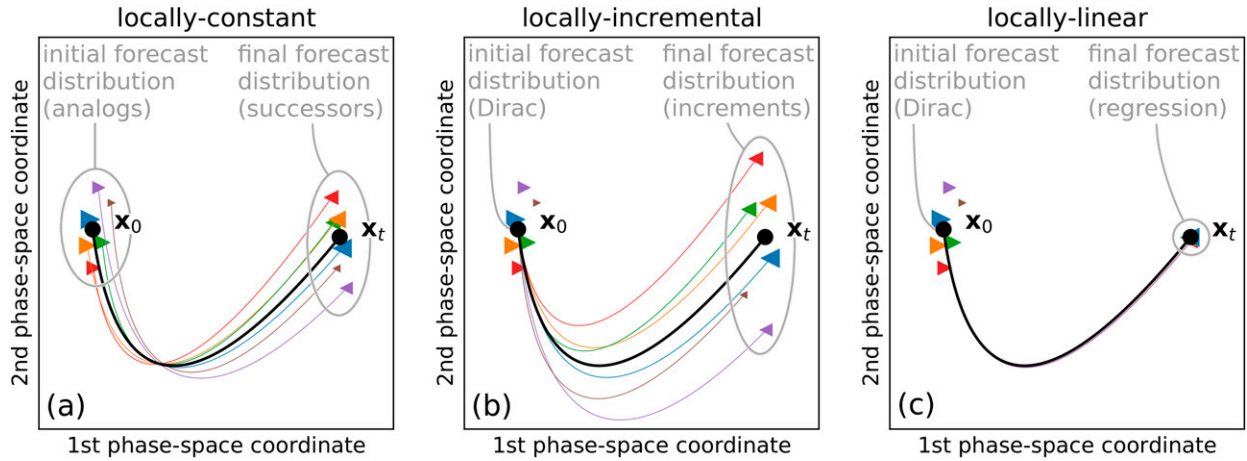


FIG. 1. Analog forecasting operators presented in section 2b. The flow map $\Phi^t(\mathbf{x}_0)$ has a simple polynomial form. Analogs are drawn from a normal distribution centered on $\mathbf{x}_0$ and follow the same model as the real state $\mathbf{x}$. The same analogs and flow maps are used for the three operators (a) LC, (b) LI, and (c) LL. Weights ω_k are computed using Gaussian kernels. The size of the k th triangle is proportional to ω_k . The real initial and future states $\mathbf{x}_0$ and $\mathbf{x}_t$ are shown in full circles. The initial forecast distribution is given either by the analogs in (a) or by a Dirac delta at $\mathbf{x}_0$ in (b) and (c). The final forecast distribution is given either by the successors in (a) or by the increments added to $\mathbf{x}_0$ in (b) or by residuals added to a linear regression applied to $\mathbf{x}_0$ in (c).

$$\mathbf{f}_a = \mathbf{f} + \delta \tilde{\mathbf{f}}, \quad (4c)$$

or in an integrated form using flow maps:

$$\mathbf{x}_t = \Phi^t(\mathbf{x}_0), \quad (5a)$$

$$\forall k, \quad \mathbf{a}_t^k = \Phi_a^t(\mathbf{a}_0^k), \quad (5b)$$

$$\Phi_a^t = \Phi^t + \delta \tilde{\Phi}^t, \quad (5c)$$

where Φ_a is the flow map of the analogs, and $\tilde{\Phi}$ is the difference between the analog and real flow maps normalized through the scalar value δ so that Φ , Φ_a , and $\tilde{\Phi}$ are of the same order of magnitude. In particular, in the following, the limit $\delta \rightarrow 0$ will be taken, equivalently to the limit $|\Phi'| \gg |\delta \tilde{\Phi}'|$. This will allow us to emphasize that at first order in δ , the contribution of the error in the analogs' flow map (or “dynamical error”) is additive and linear in the small parameter δ . The maps $\mathbf{f}$, $\mathbf{f}_a$, and $\tilde{\mathbf{f}}$ are defined accordingly.

To account for observational errors in the catalog, we use an error-in-variables framework:

$$\forall k, \quad \mathbf{a}_t^k - \boldsymbol{\epsilon}_t^k = \Phi_a^t(\mathbf{a}_0^k - \boldsymbol{\epsilon}_0^k), \quad (5d)$$

where $\boldsymbol{\epsilon}_0^k$ is the additive observational error of the analog, and $\boldsymbol{\epsilon}_t^k$ is the error of the successor. In the description using $\mathbf{f}$, the observational noises would be added at time 0 and t to a trajectory that follows the equation given by $\mathbf{f}_a$. The noises do not perturb the trajectory, only the observation.

In these formulations, the fundamental hypotheses of analog forecasting are the continuity of Φ^t (or $\mathbf{f}$) with respect to the phase-space variable $\mathbf{x}$, the density of the catalog $\mathcal{C}$ (i.e., for all k , $\mathbf{a}_0^k$ is close to $\mathbf{x}_0$ for a given metric) and the adequacy of the analogs' dynamics (i.e., δ is small, $\Phi_a \approx \Phi$).

One must assume that the catalog yields enough samples to be dense in the phase space (in the topological exception),

which means that no region of phase space is left unexplored. However, the minimum catalog length to reach a given density depends exponentially on the dimension of the system (Van Den Dool 1994; Nicolis 1998), making the use of analogs in high-dimension challenging. Also, some variables may exhibit extreme events that belong to regions of phase space that are rarely visited (distribution tails), even when the phase-space dimension is low. Therefore, one must be especially careful when using analogs to forecast extreme events.

In practice, the fact that the analogs' dynamics are close to the real dynamics of the system (i.e., $\delta \rightarrow 0$) is only partially achieved. For instance, climate change may induce some modifications in atmospheric dynamics between the preindustrial era and the twenty-first century, so that the hypothesis $\Phi_a \approx \Phi$ might not be valid if one uses analogs of wind velocity from the 1960s to produce wind velocity forecasts in 2020. Also, if one uses analogs from numerical model output, discretization approximations and physical approximations imply that $\Phi_a \neq \Phi$.

b. When analogs work: Taylor expansions of the dynamics

1) DISTANCE BETWEEN SUCCESSOR AND REAL STATE

Assuming different levels of smoothness of the flow maps and using Taylor expansions, one can estimate the difference between the real future state $\mathbf{x}_t$ and any given successor $\mathbf{a}_t^k$ at leading order:

$$\forall k, \quad \mathbf{a}_t^k - \mathbf{x}_t = \delta \tilde{\Phi}^t(\mathbf{x}_0) + [\nabla \Phi^t|_{\mathbf{x}_0}] \cdot (\mathbf{a}_0^k - \mathbf{x}_0) + \mathcal{O}(|\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta |\mathbf{a}_0^k - \mathbf{x}_0|), \quad (6a)$$

where $\nabla \Phi^t|_{\mathbf{x}_0}$ is the Jacobian matrix (the matrix of partial derivatives in phase space) of Φ^t at $\mathbf{x}_0$, “ $\cdot$ ” is the matrix multiplication, and $\mathcal{O}(|\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta |\mathbf{a}_0^k - \mathbf{x}_0|)$ represents higher-order

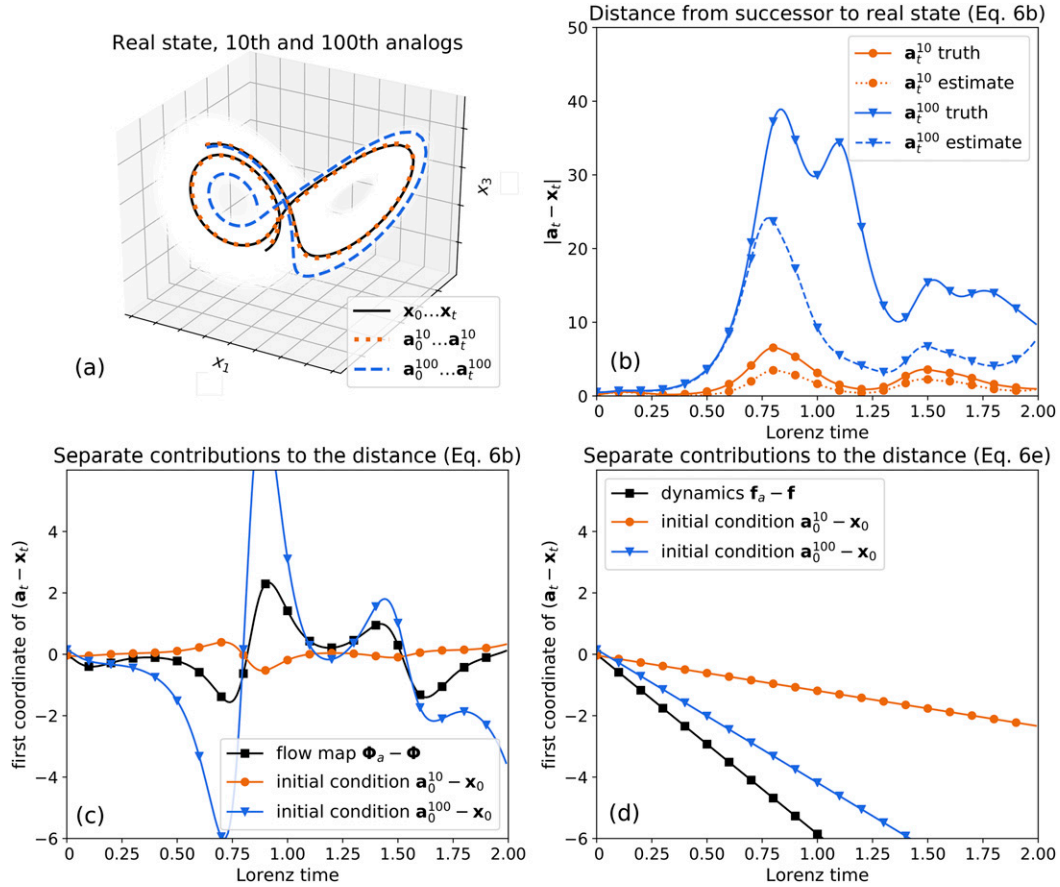


FIG. 2. Illustrating Eqs. (6b) and (6e) on the three-variable L63 system. (a) A real trajectory from $\mathbf{x}_0$ to $\mathbf{x}_t$ and two analog trajectories, namely the 10th best analog $\mathbf{a}_0^{10}$ to $\mathbf{a}_t^{10}$ and the 100th best analog $\mathbf{a}_0^{100}$ to $\mathbf{a}_t^{100}$. The catalog is shown in white. (b) Comparing the exact value of the norm of $\mathbf{a}_t - \mathbf{x}_t$ (full lines) and the sum of the two terms on the right-hand side of Eq. (6b) (dashed lines). (c) Contributions of the first term (black squares) and the second term (brown circles and blue triangles) of the right-hand side of Eq. (6b) projected on the first coordinate of the L63 system. (d) Contributions of the first term (black squares) and the second term (brown circles and blue triangles) of the right-hand side of Eq. (6e) projected on the first coordinate of the L63 system.

terms. The notation $\alpha = \mathcal{O}(\beta)$ means that the absolute ratio of α over β is bounded as β goes toward zero. We also use the notation $\mathcal{O}(\alpha, \beta) = \mathcal{O}(\alpha) + \mathcal{O}(\beta)$.

Neglecting higher-order terms and simplifying notations, Eq. (6b) can be rewritten:

$$\mathbf{a}_t^k - \mathbf{x}_t \approx \delta \tilde{\Phi}^t + \nabla \Phi^t \cdot (\mathbf{a}_0^k - \mathbf{x}_0), \quad (6b)$$

where the evaluation of $\delta \tilde{\Phi}^t$ and $\nabla \Phi^t$ at $\mathbf{x}_0$ is implicit. In the presence of observational noise as in Eq. (5d), two terms are added:

$$\mathbf{a}_t^k - \mathbf{x}_t \approx \delta \tilde{\Phi}^t + \nabla \Phi^t \cdot (\mathbf{a}_0^k - \mathbf{x}_0) + \boldsymbol{\epsilon}_t^k - \nabla \Phi^t \cdot \boldsymbol{\epsilon}_0^k. \quad (6c)$$

The leading-order difference terms explicitly described in the right-hand side of Eq. (6b) come from two sources. The first source is the difference between the analog and true flow maps at point $\mathbf{x}_0$, which is independent of $\mathbf{a}_0^k$. The second source of difference is the mismatch in the initial condition, left-multiplied by the Jacobian matrix of the true flow map at point $\mathbf{x}_0$.

Equation (6b) states that at first order, these two error terms are additive. This is not true at higher orders. Higher-order terms include the bilinear product of $\mathbf{a}_0^k - \mathbf{x}_0$ with a matrix of second derivatives of Φ^t called the Hessian, and the product of the Jacobian of $\tilde{\Phi}^t$ at $\mathbf{x}_0$ and $\mathbf{a}_0^k - \mathbf{x}_0$. Note also that, although $|\mathbf{a}_0^k - \mathbf{x}_0|^{-2} \mathcal{O}(|\mathbf{a}_0^k - \mathbf{x}_0|^2)$ is bounded, it can still be of large magnitude. In particular, for large lead times, higher-order terms can become dominant due to the chaotic divergence associated with positive Lyapunov exponents. A detailed example will be given in Fig. 7.

Figure 2 shows applications of Eq. (6b) to the three-variable system of Lorenz (1963), hereafter L63. We focus on Eq. (6b) rather than the full Eq. (6c), because $\boldsymbol{\epsilon}_t^k$ plays the same role as $\delta \tilde{\Phi}^t$ and $\boldsymbol{\epsilon}_0^k$ plays the same role as $(\mathbf{a}_0^k - \mathbf{x}_0)$. A real trajectory is compared with two analog trajectories. The L63 system is solved numerically using a fourth-order Runge–Kutta finite-difference scheme, with numerical integration time step $\Delta t = 0.01$ nondimensional time. For notation details, see Eq. (A1) in appendix A. The real trajectory has parameters $\sigma = 10, \rho = 28$,

and $\beta = 8/3$, while the σ parameter for the analog dynamics is slightly perturbed with $\tilde{\sigma} = 9 = 0.9\sigma$. The matrices $\delta\tilde{\Phi}'$ and $\nabla\Phi'$ are estimated numerically using formulas given below and time step $\Delta t = 0.01$. The 10th analog stays close enough to the real trajectory all the time (Fig. 2a), therefore Eq. (6b) gives a satisfactory approximation of $|\mathbf{a}_t^{10} - \mathbf{x}_t|$ (Fig. 2b). The 100th analog starts to be too far from the real trajectory around $t \approx 0.7$ (Fig. 2a,b), and Eq. (6b) provides a poor approximation of $|\mathbf{a}_t^{100} - \mathbf{x}_t|$ (Fig. 2b).

The different right-hand-side terms of Eq. (6b) are projected on the first axis of phase space and displayed in Fig. 2c. The “flow map” term $\delta\tilde{\Phi}'$ is the same for both analogs, but the “initial condition” term $\nabla\Phi' \cdot (\mathbf{a}_0^k - \mathbf{x}_0)$ is much larger for the 100th analog, and one can see that those terms are proportional, here negatively correlated.

Further assuming that first-order terms in t are dominant, one can express Eq. (6b) in the alternative formulation:

$$\forall k, \quad \mathbf{a}_t^k - \mathbf{x}_t = t\delta\tilde{\mathbf{f}}(\mathbf{x}_0) + [\mathbf{I} + t\nabla\mathbf{f}|_{\mathbf{x}_0}] \cdot (\mathbf{a}_0^k - \mathbf{x}_0) + \mathcal{O}(t^2, |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta|\mathbf{a}_0^k - \mathbf{x}_0|), \quad (6d)$$

where $\mathbf{I}$ is the identity matrix. Using lighter notations, this becomes

$$\mathbf{a}_t^k - \mathbf{x}_t \approx t\delta\tilde{\mathbf{f}} + [\mathbf{I} + t\nabla\mathbf{f}] \cdot (\mathbf{a}_0^k - \mathbf{x}_0), \quad (6e)$$

where the evaluation of $\delta\tilde{\mathbf{f}}$ and $\nabla\mathbf{f}$ at $\mathbf{x}_0$ is implicit. This last formulation is analogous to a Euler scheme used in finite-difference numerical methods for solving differential equations. It allows us to describe the behavior of analog forecasting for small times, before the divergence associated with positive Lyapunov exponents becomes dominant. In Fig. 2d, one can see that the right-hand-side terms of Eq. (6e) only approximate the terms of Eq. (6b) for $t \leq 0.1$. This also explains the behavior of some analog forecasting operators in Fig. 7 (introduced later on in the paper) for $t \leq 0.1$.

2) LINK BETWEEN THE TWO FORMULATIONS, $\mathbf{f}$ AND Φ'

Equation (6e) is a first-order expansion in time of Eq. (6b). The fundamental resolvent matrix $\mathbf{M}(t, t')$ gives a more complete relationship between the two representations. $\mathbf{M}(t, t')$ is solution to the time varying linear system $d\mathbf{M}(t, t')/dt = \nabla\mathbf{f}|_{\mathbf{x}_t} \cdot \mathbf{M}(t, t')$ with $\mathbf{M}(t', t') = \mathbf{I}$. The fundamental resolvent matrix can be approximated numerically as $\mathbf{M}(t, t') \approx \exp(\Delta t \nabla\mathbf{f}_t) \cdot \exp(\Delta t \nabla\mathbf{f}_{t-\Delta t}) \dots \exp(\Delta t \nabla\mathbf{f}_{t'})$ with numerical time step Δt and using the short notation $\nabla\mathbf{f}_t := \nabla\mathbf{f}|_{\mathbf{x}_t}$.

We have

$$\delta\tilde{\Phi}'(\mathbf{x}_0) \approx \delta \int_0^t \mathbf{M}(t, t') \cdot \tilde{\mathbf{f}}(\mathbf{x}_{t'}) du, \quad (7a)$$

$$\nabla\Phi'|_{\mathbf{x}_0} = \mathbf{M}(t, 0), \quad (7b)$$

where the “ $\approx$ ” sign is here to say that Eq. (7a) is valid only at first order in δ . This first order is enough to compute the right-hand-side terms of Eq. (6b), which is also valid at first order in δ .

From Eq. (7b), one can use Taylor developments relating $\nabla\mathbf{f}$ and $\nabla\Phi'$, such as

$$\nabla\Phi' = \mathbf{I} + t\nabla\mathbf{f}_0 + t^2 \left[(\nabla\mathbf{f}_0)^2 + \frac{d}{dt} \nabla\mathbf{f}_0 \right] + \mathcal{O}(t^3), \quad (8)$$

where $\nabla\Phi'$ is implicitly evaluated at $\mathbf{x}_0$. The short notation $\nabla\mathbf{f}_0$ is used for $\nabla\mathbf{f}|_{\mathbf{x}_0}$, and $(d/dt)\nabla\mathbf{f}_0$ is the time derivative along the trajectory $\mathbf{x}_t$ of the Jacobian of $\mathbf{f}$, at $t = 0$; $(d/dt)\nabla\mathbf{f}|_0 := \lim_{t \rightarrow 0} (\nabla\mathbf{f}_t - \nabla\mathbf{f}_0)/t$. At first order in t , one recovers the result expressed in Eq. (6e).

4. Consequences for analog forecasting operators

a. Mean error of analog forecasting operators

By multiplying Eqs. (6a) and (3b) by ω_k and summing over k , one can compare the distances from $\mathbf{x}_t$ to the averages μ_{LC} , μ_{LI} , and μ_{LL} of the different analog forecasting operators of section 2b. Those averages depend on t , although only implicitly in the notation. Let $\mu_0 = \sum_k \omega_k \mathbf{a}_0^k$ be the weighted mean of the analogs, we then have the following expressions.

LC mean error:

$$\begin{aligned} \mu_{LC} - \mathbf{x}_t &= \delta\tilde{\Phi}'(\mathbf{x}_0) + [\nabla\Phi'|_{\mathbf{x}_0}] \cdot (\mu_0 - \mathbf{x}_0) \\ &+ \mathcal{O}\left(\sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right), \end{aligned} \quad (9a)$$

$$\begin{aligned} \mu_{LC} - \mathbf{x}_t &= t\delta\tilde{\mathbf{f}}(\mathbf{x}_0) + [\mathbf{I} + t\nabla\mathbf{f}|_{\mathbf{x}_0}] \cdot (\mu_0 - \mathbf{x}_0) \\ &+ \mathcal{O}\left(t^2, \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right). \end{aligned} \quad (9b)$$

Using simpler notations with implicit evaluation at $\mathbf{x}_0$, this gives

$$\mu_{LC} - \mathbf{x}_t \approx \delta\tilde{\Phi}' + \nabla\Phi' \cdot (\mu_0 - \mathbf{x}_0), \quad (9c)$$

$$\approx t\delta\tilde{\mathbf{f}} + [\mathbf{I} + t\nabla\mathbf{f}] \cdot (\mu_0 - \mathbf{x}_0), \quad (9d)$$

and, in the presence of noise:

$$\mu_{LC} - \mathbf{x}_t \approx \delta\tilde{\Phi}' + \nabla\Phi' \cdot (\mu_0 - \mathbf{x}_0) + \sum_k \omega_k \epsilon_t^k - \nabla\Phi' \cdot \sum_k \omega_k \epsilon_0^k. \quad (9e)$$

LI mean error:

$$\begin{aligned} \mu_{LI} - \mathbf{x}_t &= \delta\tilde{\Phi}'(\mathbf{x}_0) + [\nabla\Phi'|_{\mathbf{x}_0} - \mathbf{I}] \cdot (\mu_0 - \mathbf{x}_0) \\ &+ \mathcal{O}\left(\sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right), \end{aligned} \quad (10a)$$

$$\begin{aligned} \mu_{LI} - \mathbf{x}_t &= t\delta\tilde{\mathbf{f}}(\mathbf{x}_0) + [t\nabla\mathbf{f}|_{\mathbf{x}_0}] \cdot (\mu_0 - \mathbf{x}_0) \\ &+ \mathcal{O}\left(t^2, \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right). \end{aligned} \quad (10b)$$

Using simpler notations with implicit evaluation at $\mathbf{x}_0$, this gives

$$\mu_{LI} - \mathbf{x}_t \approx \delta\tilde{\Phi}' + [\nabla\Phi' - \mathbf{I}] \cdot (\mu_0 - \mathbf{x}_0), \quad (10c)$$

$$\approx t\delta\tilde{\mathbf{f}} + t\nabla\mathbf{f} \cdot (\mu_0 - \mathbf{x}_0). \quad (10d)$$

And, in the presence of noise,

$$\boldsymbol{\mu}_{\text{LI}} - \mathbf{x}_t \approx \delta \tilde{\Phi}' + [\nabla \Phi' - \mathbf{I}] \cdot (\boldsymbol{\mu}_0 - \mathbf{x}_0) + \sum_k \omega_k \boldsymbol{\epsilon}_t^k - \nabla \Phi^t \cdot \sum_k \omega_k \boldsymbol{\epsilon}_0^k. \quad (10e)$$

The errors of the LC and LI operators are both affected by the difference between the analog and real flow maps. This source of error cannot be circumvented unless provided with some information about $\delta \tilde{\Phi}$. The other first-order error term is linear in $(\boldsymbol{\mu}_0 - \mathbf{x}_0)$, but when $t \rightarrow 0$, this term is of order t in the LI case. Thus, for small lead-times (i.e., when the first-order time approximations are valid), as both $t \rightarrow 0$ and $\boldsymbol{\mu}_0 \rightarrow \mathbf{x}_0$ (dense catalog), the mean of the LI provides a better estimate of $\mathbf{x}_t$. This is why this operator is qualified by Lguensat et al. (2017) as more “physically sound” than the LC: the LI takes advantage of the fact that $\lim_{t \rightarrow 0} \Phi^t = \mathbf{I}$, just as any finite-difference numerical scheme does. Formulas similar to Eqs. (9) and (10) were used by Platzter et al. (2019) to predict analog forecasting errors with LC and LI operators, on the famous three-variable L63 system, with $\delta = 0$.

Another interesting property of the LI is that it can give estimates of $\mathbf{x}_t$ out of the convex hull of the catalog. This is related to what is called “novelty creation” in the machine-learning terminology. Such a property is interesting, because it could help to generalize out of what has already been observed. Since observations never span the whole phase space, it is useful to be able to produce forecasts beyond past observations. However, this can also generate some inconsistent forecasts. Indeed, if t is not small enough, the LI operator can produce forecasts that have a large error due to the $-\mathbf{I}$ term in Eq. (10a). In Fig. 1, one can see that the LI has a larger variance than the LC for large times.

In Eq. (9e) and Eq. (10e), the noises $\boldsymbol{\epsilon}_0^k$ and $\boldsymbol{\epsilon}_t^k$ are averaged with weights ω_k . This averaging procedure gives new centered variables $\sum_k \omega_k \boldsymbol{\epsilon}_0^k$ and $\sum_k \omega_k \boldsymbol{\epsilon}_t^k$. If the noises have finite variance σ^2 , the variance of the averaged noises is $\approx \sigma^2/K$ [to be more precise, the variance is divided by $\approx K/2$ if one uses the procedure of Lguensat et al. (2017)]. This shows that there is a bias–variance trade-off: raising the value of K lowers the variance and raises the bias of the estimator of $\mathbf{x}_t$.

Equation (9) is also valid for constructed analogs (CA) introduced in section 4b, where the weights $\{\omega_k^{\text{CA}}\}$ are chosen so that $|\sum_k \omega_k^{\text{CA}} \mathbf{a}_0^k - \mathbf{x}_0|$ is as small as possible. This means that the $(\boldsymbol{\mu}_0 - \mathbf{x}_0)$ -linear term of Eq. (9) is also small. As mentioned earlier, Tippet and DelSole (2013) showed that this strategy is equivalent to making a linear regression. This explains why the $(\boldsymbol{\mu}_0 - \mathbf{x}_0)$ -linear term is absent from Eqs. (11a) and (11b). LI mean error:

$$\boldsymbol{\mu}_{\text{LL}} - \mathbf{x}_t = \delta \tilde{\Phi}'(\mathbf{x}_0) + \mathcal{O}\left(\sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right), \quad (11a)$$

$$\boldsymbol{\mu}_{\text{LI}} - \mathbf{x}_t = t \delta \tilde{\mathbf{f}}(\mathbf{x}_0) + \mathcal{O}\left(t^2, \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right). \quad (11b)$$

Another way to understand why the $(\boldsymbol{\mu}_0 - \mathbf{x}_0)$ -linear term should disappear when using the LL is to see that the LL is estimating the local Jacobian of the flow map. Indeed, the linear regression between the analogs and the successors gives an estimation of $\nabla \Phi'|_{\mathbf{x}_0}$, with an estimation error that is at least of order $\mathcal{O}(|\boldsymbol{\mu}_0 - \mathbf{x}_0|, \delta)$. Section 4b gives a detailed argumentation to support this claim and investigates limitations. The estimation error between the linear regression matrix and the Jacobian thus adds higher-order error terms to the right-hand side of Eqs. (11a) and (11b), but these are already included in the $\mathcal{O}\left(\sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2, \delta \sum_k \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|\right)$.

The modified form of Eqs. (11a) and (11b) due to additive observational noise is not straightforward. The linear regression of the LL operator is designed for the special case of $\boldsymbol{\epsilon}_t^k \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\boldsymbol{\epsilon}_0^k = \mathbf{0}$. The error-in-variables model where $\boldsymbol{\epsilon}_t^k, \boldsymbol{\epsilon}_0^k \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ is a more realistic representation of measurement errors. Applying regular linear least squares regression in this case is known to induce bias in the estimated linear coefficients. The bias is toward zero when Φ^t is one-dimensional and linear, but it is more complicated when Φ^t is nonlinear (Griliches and Ringstad 1970) and multidimensional. Total least squares (Markovsky and Van Huffel 2007) could be used to account for this fact. However, for the sake of simplicity here, the LL operator presented in section 3b is applied. A numerical example will be described in section 4b(5), section 4c and in Fig. 7.

We now make the explicit link between the three operators. Recall the notations of section 2b: the LL operator finds slope $\mathbf{S}$ and intercept $\mathbf{c}$ such that for all k , $\mathbf{a}_t^k = \mathbf{S}(\mathbf{a}_0^k - \boldsymbol{\mu}_0) + \mathbf{c} + \boldsymbol{\xi}^k$ using weighted least squares estimates. This gives $\mathbf{c} = \sum_k \omega_k \mathbf{a}_t^k = \boldsymbol{\mu}_{\text{LC}}$, thus we have $\boldsymbol{\mu}_{\text{LL}} = \boldsymbol{\mu}_{\text{LC}} + \mathbf{S}(\mathbf{x}_0 - \boldsymbol{\mu}_0)$ and the following relations hold:

$$\boldsymbol{\mu}_{\text{LC}} = \boldsymbol{\mu}_{\text{LL}}|_{\mathbf{S}=\mathbf{0}}, \quad (12a)$$

$$\boldsymbol{\mu}_{\text{LI}} = \boldsymbol{\mu}_{\text{LL}}|_{\mathbf{S}=\mathbf{I}}, \quad (12b)$$

such that the LC and LI operator are particular cases of the LL operator. We also have $\lim_{t \rightarrow 0} \mathbf{S} = \mathbf{I}$, because for all k , $\lim_{t \rightarrow 0} \mathbf{a}_t^k = \mathbf{a}_0^k$. Thus, mean forecasts of the LL and LI operators are equivalent as t approaches 0: $\boldsymbol{\mu}_{\text{LL}} \xrightarrow{t \rightarrow 0} \boldsymbol{\mu}_{\text{LI}}$.

This analysis shows that, for small lead times (i.e., when the time linear approximation is still valid) and in terms of mean forecast error, the LL operator is more accurate than the LI, and the latter is more accurate than the LC. These findings are in agreement with the numerical experiments of Lguensat et al. (2017). A detailed description of the advantages of each analog forecasting operator for different lead times is given in section 4c, in a numerical experiment of the L63 system.

We now investigate the link between the local Jacobian of the flow $\nabla \Phi'|_{\mathbf{x}_0}$, and the linear regression matrix from the LL operator $\mathbf{S}$.

b. Ability of analogs to estimate local Jacobians

If analogs can estimate the Jacobian of the real system, it means that analog forecasting provides a local approximation of the real dynamics, proving the relevance of analogs for short-range forecasts. Furthermore, having an estimation of the local Jacobian can be useful in some applications such as the extended Kalman filter (see Jazwinski 1970 for a detailed

introduction), where the Jacobian allows us to estimate the evolution of the forecast covariance.

1) DERIVATION OF THE FIRST-ORDER ERROR IN JACOBIAN ESTIMATION

It is possible to find an exact expression of the first-order error term in the estimation of the local Jacobian. Let us start with the case of perfect agreement between the real and analog flow maps: $\Phi_a \approx \Phi$, or $\delta = 0$. Then, assume that in the neighborhood of $\mathbf{x}_0$ where the analogs lie, the flow $\Phi^t(\cdot)$ can be approximated by a quadratic function in phase space. We then have

$$\forall k, \quad \mathbf{a}_t^k = \nabla \Phi^t(\mathbf{a}_0^k - \boldsymbol{\mu}_0) + \frac{1}{2}(\mathbf{a}_0^k - \boldsymbol{\mu}_0) \nabla^2 \Phi^t(\mathbf{a}_0^k - \boldsymbol{\mu}_0)^T + \text{Cst}, \quad (13)$$

where “Cst” is a constant (independent of k), and the Jacobian and Hessian of Φ^t are implicitly evaluated at $\mathbf{x}_0$ (see [appendix B](#) for notation of product of vectors and Hessian). In the next equations, the t superscript is dropped to simplify notations. Let $\mathbf{X}$ be the matrix of the analogs minus their mean, so that the k th row of $\mathbf{x}$ is $\mathbf{a}_0^k - \boldsymbol{\mu}_0$. Similarly, let $\mathbf{Y}$ be the matrix of the successors, with the k th row of $\mathbf{Y}$ being $\mathbf{a}_t^k$. Equation (13) thus translates into $\mathbf{Y} = \mathbf{X} \nabla \Phi + (1/2) \mathbf{X} \nabla^2 \Phi \mathbf{X}^T$, omitting the constant.

Now let $\Omega = \text{diag}(\omega_1, \dots, \omega_K)$ be the $K \times K$ diagonal matrix of the weights given to each analog in the regression. Then $\mathbf{S}$ is the weighted least squares solution of the linear regression $\mathbf{S} = (\mathbf{X}^T \Omega \mathbf{X})^{-1} \mathbf{X}^T \Omega \mathbf{Y}$. With a bit of rewriting, this finally gives

$$\mathbf{S} - \nabla \Phi = (\mathbf{X}^T \Omega \mathbf{X})^{-1} \mathbf{X}^T \Omega \left\{ \mathbf{X} \nabla^2 \Phi \left[\frac{1}{2} \mathbf{X} + (\boldsymbol{\mu}_0 - \mathbf{x}_0)^T \otimes \mathbf{J}_{K,1} \right]^T \right\}, \quad (14)$$

where $\otimes$ is the Kronecker matrix product and $\mathbf{J}_{K,1}$ is the column vector with K elements all equal to 1.

Equation (14) tells us that $\mathbf{S}$ is close to the Jacobian at $\mathbf{x}_0$ up to a factor that is linear in the distance between the mean of the analogs $\boldsymbol{\mu}_0$ and the analogs $\mathbf{a}_0^k$, and another factor linear in the distance between $\boldsymbol{\mu}_0$ and $\mathbf{x}_0$. These linear error term depend on the second-order phase-space derivatives of Φ at the point $\mathbf{x}_0$ (the Hessian of Φ).

Conducting the same derivation but relaxing the hypothesis of $\delta = 0$, one would find the same result with an added linear error term involving the Jacobian of $\tilde{\Phi}$. This analysis allows us to say that $\mathbf{S} = \nabla \Phi^t + \mathcal{O}(|\boldsymbol{\mu}_0 - \mathbf{x}_0|, \delta)$, if the distance between the analogs and their mean is of same order as the distance between their mean and $\mathbf{x}_0$.

However, the claim that the linear regression matrix $\mathbf{S}$ is able to approximate the Jacobian $\nabla \Phi^t$ must be tempered by several facts. To illustrate these, the regular LL analog forecasting operator will now be compared with two other strategies aimed at solving dimensionality issues.

2) STRATEGIES FOR LINEAR REGRESSION IN HIGH DIMENSION

Dimensionality can make analog forecasting difficult, especially when using the locally linear analog forecasting operator.

Here are recalled two strategies that can be used to circumvent this issue.

The first approach uses empirical orthogonal functions (EOFs, also called principal component analysis) at every forecast step. Dimension is reduced by keeping only the first n^{eof} EOFs of the set of analogs $(\mathbf{a}_0^k)_{k \in [1,K]}$, or keeping only the n^{eof} first principal components of the matrix $\mathbf{X}^T \Omega \mathbf{X}$.

Reducing dimension using EOFs:

- Find analogs $(\mathbf{a}_0^k)_{k \in [1,K]}$ of the initial state $\mathbf{x}_0$.
- Compute the n EOFs of the weighted set of analogs $(\mathbf{a}_0^k)_{k \in [1,K]}$.
- Keep the n^{eof} first EOFs up to 95% total variance.
- Project $\mathbf{x}_0$, $(\mathbf{a}_0^k)_{k \in [1,K]}$ and $(\mathbf{a}_t^k)_{k \in [1,K]}$ on the n^{eof} first EOFs.
- Perform LL analog forecasting in this projected space.

The second strategy is to perform n analog forecasts, one for each coordinate of the phase space $\mathcal{P}$, and to assume that the future of a given coordinate only depends on the initial values of the neighboring coordinates and not on the whole initial vector $\mathbf{x}_0$. In the model of [Lorenz \(1996, hereafter L96\)](#), Eq. (A2) in appendix motivates the choice of keeping only the initial coordinates $\{i-2, i-1, i, i+2\}$ to estimate the i th future coordinate. Thus we keep only $n^{\text{trunc}} = 5$ initial coordinates. Thus, the LL operator performs n linear regressions with 5 coefficients at each forecast. By combining the results of those linear regressions, one finds an $n \times n$ matrix that is sparse by construction: all elements two cells away from the diagonal are equal to zero. This was introduced in [Lguensat et al. \(2017\)](#) as “local analogs.” In the present paper this strategy will rather be termed as “coordinate-by-coordinate” analog forecasting. We make this choice to avoid confusions between locality in phase space and locality in physical space.

Coordinate-by-coordinate forecast:

for i from 1 to n , forecast the i th future coordinate $\mathbf{x}_{t,i}$:

- Condition the forecast $\Theta_{\text{LL},i}^t$ on a few initial coordinates around $\mathbf{x}_{0,i}$.

$$\Theta_{\text{LL},i}^t(\mathbf{x}_0) = \Theta_{\text{LL},i}^t(\mathbf{x}_{0,i-2}, \mathbf{x}_{0,i-1}, \mathbf{x}_{0,i}, \mathbf{x}_{0,i+1}, \mathbf{x}_{0,i+2}).$$

- Find analogs of the truncated initial vector $(\mathbf{x}_{0,i-2}, \mathbf{x}_{0,i-1}, \mathbf{x}_{0,i}, \mathbf{x}_{0,i+1}, \mathbf{x}_{0,i+2})$.
- Perform LL analog forecasting $\Theta_{\text{LL},i}^t$.
- Store the coefficients of the linear regression $(\mathbf{S}_{i,i-2}, \mathbf{S}_{i,i-1}, \mathbf{S}_{i,i}, \mathbf{S}_{i,i+1}, \mathbf{S}_{i,i+2})$.
- Aggregate the coefficients into the $n \times n$ matrix $\mathbf{S}$

There are other possibilities than EOFs or coordinate-by-coordinate forecast to reduce dimensionality (such as dynamic mode decomposition, [Schmid 2010](#)). However, they often require prior knowledge on the system itself, so that a use on one particular set of data might not be transposable to another set without tests that are beyond the scope of this paper, as stated in the “no free lunch theorem” ([Wolpert 1996](#)). Thus, we prefer to keep an EOF analysis, as the variables we consider have a roughly Gaussian distribution, with no particularly heavy tails. The coordinate-by-coordinate forecast is justified by the nature of the L96 dynamical equations and the experiments of [Lguensat et al. \(2017\)](#). However, in the case of a real application problem (e.g., in a large spatial domain),

the coordinate-by-coordinate forecast procedure has to be carefully tuned. Indeed, the coordinates taken into account must be of the order of magnitude of the spatial correlation length.

The next section investigates limitations to the claim that the matrix $\mathbf{S}$ from the LL operator is able to approximate the Jacobian $\mathbf{V}\Phi'$, and studies the impact of dimension reduction techniques on this Jacobian estimation.

3) EFFECT OF THE NUMBER OF ANALOGS AND THE PHASE-SPACE DIMENSION

First, to be able to compute $\mathbf{S}$, one must have enough analogs to perform the inversion of the matrix $\mathbf{X}^T\mathbf{\Omega}\mathbf{X}$, where $\mathbf{X}$ is the matrix of the analogs and $\mathbf{\Omega}$ the diagonal matrix of the weights. This cannot be done unless K , the number of analogs used for the forecast, is superior or equal to n , the phase-space dimension. Using the EOF or coordinate-by-coordinate strategies from the previous section, one can reduce the dimension to n^{eof} or n^{trunc} , needing only to satisfy $K \geq n^{\text{eof}}$ or $K \geq n^{\text{trunc}}$.

To illustrate the practical consequences of these issues, numerical simulations of the L96 system were performed with $n = 8$. The L96 is a well-known chaotic dynamical system with a flexible dimension, which allows us to set the latter to a convenient value. A dimension of 8 is sufficiently high to reveal the effects of dimensionality, and low enough to allow us to display full 8×8 matrices. The governing equations were solved using a fourth-order Runge–Kutta numerical scheme with an integration time step $\Delta t = 0.05$. A catalog was built from one long trajectory (10^4 times) using the real equations ($\delta = 0$). Then, analog forecasting was performed at lead time 0.05, using the LL operator on 2×10^4 test points (10^3 nondimensional times) taken from another trajectory on the attractor (independent from the catalog). Setting the number of analogs to the limiting case $K = 9$ implies that there are just enough analogs to perform the linear regression (plus one extra analog). Even though $n = 8$ is not a very large dimension, if one is provided only with 9 good analogs, one must consider dimension reduction. Regular LL analog forecasting was compared with the combination of analog forecasting with EOFs, keeping the EOFs up to 95% variance, and with the coordinate-by-coordinate analog forecasting, with $n^{\text{trunc}} = 5$.

For a comparison with the actual atmospheric circulation, we can consider the study of Faranda et al. (2017). The authors estimate the instantaneous dimension of sea level pressure fields spanning the North Atlantic and Europe (22.5°–70°N, 80°W–50°E), with various horizontal resolutions (0.75°, 1.5°, and 2.5°) from reanalysis data. The average dimension is close to 13, with variations between 5 and 17 (25th and 75th percentiles). For comparison with our L96 example, assume that one searches for analogs of sea level pressure fields over the North Atlantic, after projection on a subspace of dimension 15. In such a case, the example that we study in this section corresponds to the situation where only 14 good analogs can be found.

To compare the matrix $\mathbf{S}$ from the linear regression between the analogs and successors and the Jacobian matrix $\mathbf{V}\Phi'$, we use the RMSE, which we define as

$$\text{RMSE}(\mathbf{S} - \mathbf{V}\Phi') = \left(\frac{1}{N_{\text{coeff}}} \sum_{i,j} (\mathbf{S}_{i,j} - \mathbf{V}\Phi'_{i,j})^2 \right)^{1/2},$$

where N_{coeff} is the number of coefficients of the matrix. For instance, if the phase-space dimension is n , then $N_{\text{coeff}} = n^2$. Thus, our definition of the RMSE between matrices is proportional to the Froebenius matrix norm of the difference. Tests were performed using the operator norm instead of the RMSE, giving similar results.

The EOF strategy ensures that the linear regression can be performed, as it projects the phase space $\mathcal{P}$ onto the EOFs that maximize the variance in the set of analogs. Thus the rank of the set of analogs is likely to be equal to n^{eof} in this reduced space. However, the EOF strategy necessarily misses some of the components of the full $n \times n$ Jacobian matrix $\mathbf{V}\Phi'$, as it gives only the estimation of an $n^{\text{eof}} \times n^{\text{eof}}$ matrix. The coordinate-by-coordinate method also ensures that the linear regression can be performed as long as n^{trunc} is low enough, but also misses some of the elements of the Jacobian matrix of the flow map. Indeed, even though the coefficients of $\mathbf{V}\mathbf{f}$ are zero two cells away from the diagonal, this is not the case of $\mathbf{V}\Phi'$. Recall that, at second-order in time, $\mathbf{V}\Phi' = \mathbf{I} + t\mathbf{V}\mathbf{f} + t^2[(\mathbf{V}\mathbf{f})^2 + (d/dt)\mathbf{V}\mathbf{f}]$. Thus, some coefficients of order t^2 will not be captured by the linear regression matrix $\mathbf{S}$ using coordinate-by-coordinate analog forecasting with $n^{\text{trunc}} = 5$.

The linear regression matrix $\mathbf{S}$ is then compared with $\mathbf{V}\Phi'$ for the three methods. The real value of $\mathbf{V}\Phi'$ is estimated with the second-order time expansion of Eq. (8) that can be computed directly from the model Eq. (A2). An example is shown in Fig. 3. In this case, the regular analog forecasting misses the Jacobian with RMSE of 0.317, because the rank of the set of analogs is too low and $\mathbf{X}^T\mathbf{\Omega}\mathbf{X}$ is thus not invertible. Analog forecasting combined with EOFs gives a better result as it circumvents this inversion problem, with a total RMSE between $\mathbf{S}$ and $\mathbf{V}\Phi'$ of 0.151. The coordinate-by-coordinate analog forecasting gives the best solution in this case, with a RMSE of 0.053. Note that many coefficients of the matrix $\mathbf{S}$ are set to zero by construction when using the coordinate-by-coordinate method.

Then, Fig. 4 shows empirical probability density functions for the RMSE of $\mathbf{S} - \mathbf{V}\Phi'$ for each of the three methods. The low number of analogs implies large fluctuation of the regular LL analog forecasts, as the rank of the set of analogs used can be below or close to the phase-space dimension, making the inversion of $\mathbf{X}^T\mathbf{\Omega}\mathbf{X}$ hazardous. This variability is noticeably reduced when the inversion is performed in the n^{eof} -dimension reduced space. The EOF strategy has the advantage of preventing large errors and the drawback of hindering accurate estimations of the Jacobian [$\log_{10}(\text{RMSE}) < -1$]. Indeed, when using EOFs the linear regression matrix has a rank necessarily lower than n , and some information is missed. Finally, coordinate-by-coordinate analog forecasting is able to perform better estimations of the Jacobian in average, and with a variability between that of the regular analogs and that of the analogs combined with EOFs. However, the probability to have accurate estimations of the Jacobian [$\log_{10}(\text{RMSE}) < -1.4$] is lower with coordinate-by-coordinate

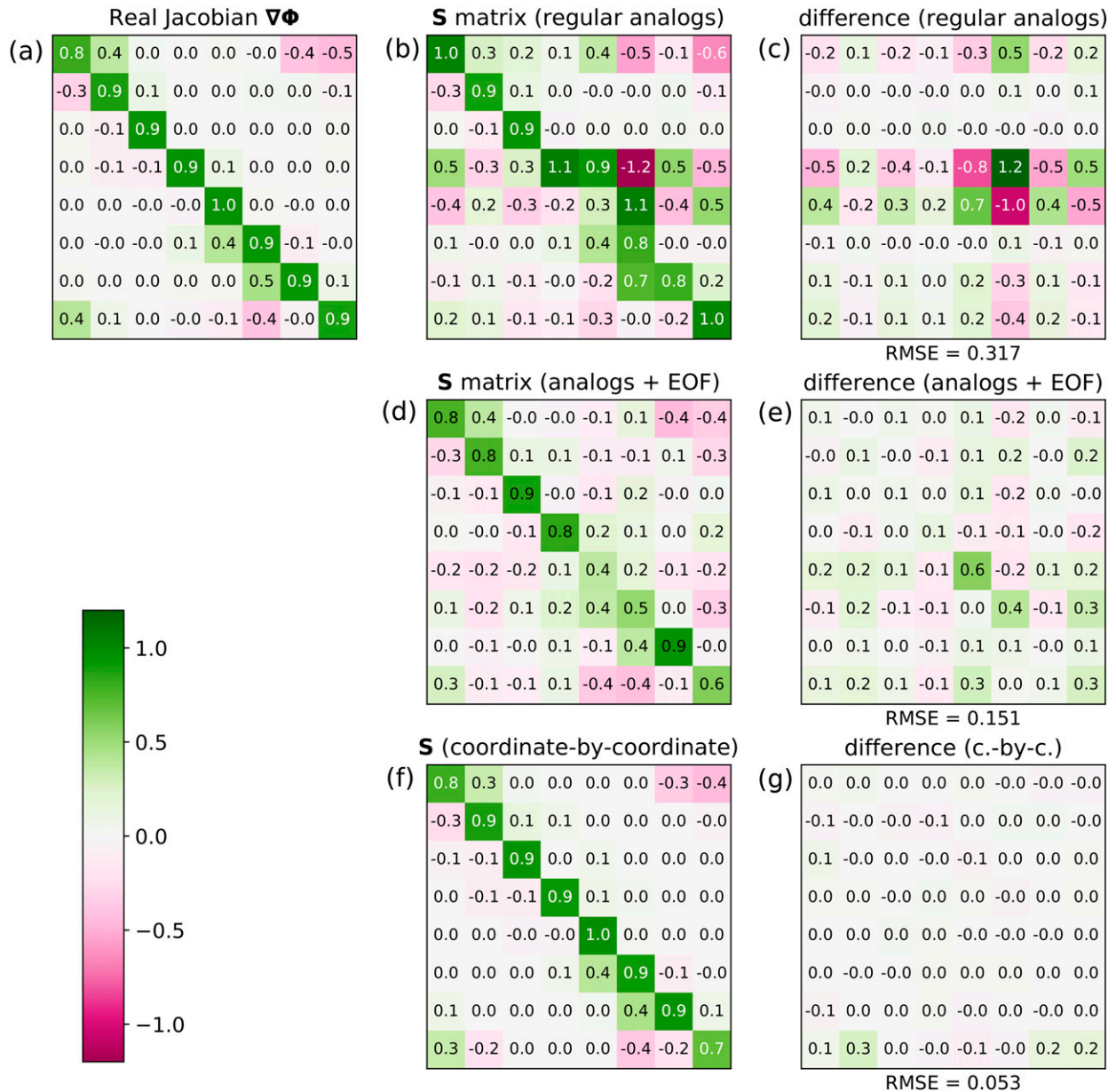


FIG. 3. Flow map Jacobian matrix estimation with the model of Lorenz (1996), without observational noise. Forecast lead time is $t = 0.05$ Lorenz time, catalog length is 10^4 Lorenz times, phase-space dimension is $n = 8$. $K = 9$ analogs are used for the forecast and Gaussian kernels for the weights ω_k with shape parameter λ set to the median of analog-to-state distances $|\mathbf{a}_0^k - \mathbf{x}_0|$. (a) Jacobian matrix $\nabla\Phi|_{\mathbf{x}_0}$. (b) Linear regression matrix $\mathbf{S}$ using regular analogs. (c) Difference $\mathbf{S} - \nabla\Phi|_{\mathbf{x}_0}$ with regular analogs, also giving the value of RMSE below the plot. (d),(e) As in (a)–(c), but the linear regression is performed in a lower-dimensional subspace spanned by the first EOFs of the set of the $K = 9$ analogs. (f),(g) As in (d) and (e), but the linear regression is performed coordinate by coordinate, and assuming that the coefficients are zero two cells away from the diagonal.

analog forecasting than with regular analog forecasting. This can be witnessed as the area under the graph for $\log_{10}(\text{RMSE}) < -1.4$ is larger for regular analogs than for coordinate-by-coordinate analog forecasting. This is due to the small (order t^2) nonzero coefficients two cells away from the diagonal that the coordinate-by-coordinate analog forecasting cannot estimate.

In some situations, however, the number of analogs K is much larger than the phase-space dimension n , but the linear

regression matrix $\mathbf{S}$ is still unable to approximate the Jacobian $\nabla\Phi$. The next section gives an example of such a situation.

4) EFFECT OF THE ANALOGS' RANK AND THE ATTRACTOR'S DIMENSION

As we have seen, to calculate $\mathbf{S}$ and perform LL analog forecasting, one must invert the matrix $\mathbf{X}^T\mathbf{O}\mathbf{X}$. This means that the set of analogs must be of rank n . Yet, in some situations, the

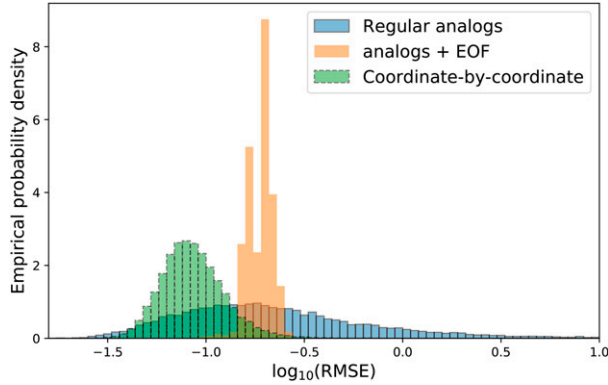


FIG. 4. Empirical probability density function of RMSE in flow map Jacobian matrix estimation, depending on the method used. We use the system of Lorenz (1996) with phase-space dimension $n = 8$. $K = 9$ analogs are used for each forecast and the methods are as in Fig. 3.

dimension of the attractor is lower than the full phase-space dimension n . Thus, if the catalog is made of one trajectory inside the attractor, the set of analogs might not be of rank n , however large K might be. In some cases, the dimension of the attractor is between $n - 1$ and n , such that the matrix $\mathbf{X}^T \mathbf{\Omega} \mathbf{X}$ is still invertible but very sensitive to fluctuations in the rank of the set of analogs.

Similar remarks can be made for the successors. If $\mathbf{Y}$ (the set of successors) is not of rank n , then the matrix $\mathbf{S}$, if it can be computed, is still not of rank n . Thus $\mathbf{S}$ will not be able to estimate the Jacobian $\mathbf{V}\Phi'$ if the latter is of rank n . Note that the rank of the successors (the rank of the matrix $\mathbf{Y}$) is highly dependent on the rank of the analogs and the Jacobian matrix as we have $\mathbf{Y} \approx \mathbf{X}\mathbf{V}\Phi'$ at first order in $\mathbf{X}$, such that if the analogs are not of rank n the successors are likely not to be of rank n either.

Thus, depending on the dimension of the attractor, the LL analog forecasting operator might not be able to estimate the local Jacobian of the real flow map, but only a projection of this Jacobian matrix onto the local sets of analogs and successors. This is a typical case where data-driven methods are not able to reveal the full physics of an observed system unless provided with other sources of information or hypotheses, such as a parametric law.

The three-variable L63 system is used to illustrate this fact. This system is known to have a dimension of ≈ 2.06 , with local variations around this value (Caby et al. 2019). This is the perfect case study where the rank of the set of analogs will be close to $n - 1$. Thus, the linear regression matrix $\mathbf{S}$ between the analogs and the successors is not able to approximate the full (3×3) Jacobian matrix $\mathbf{V}\Phi'$. Using restriction to the vector subspace V_a spanned by the two first EOFs of the analogs ($\mathbf{e}_1^a, \mathbf{e}_2^a$), one can understand better the connection between the two matrices $\mathbf{V}\Phi'$ and $\mathbf{S}$. In the following, subscript r indicates restriction to ($\mathbf{e}_1^a, \mathbf{e}_2^a$). The choice of using only the two first EOFs is motivated by the quasi-planar nature of the Lorenz attractor. In the next formulas the r superscript is dropped for the sake of readability:

$$\mathbf{V}\Phi_r = \mathbf{V}\Phi \begin{pmatrix} \mathbf{e}_1^a \\ \mathbf{e}_2^a \\ 0 \end{pmatrix}, \quad (15a)$$

$$\mathbf{S}_r = \mathbf{S} \begin{pmatrix} \mathbf{e}_1^a \\ \mathbf{e}_2^a \\ 0 \end{pmatrix}. \quad (15b)$$

The condition number of the set of analogs gives a direct way to measure whether the matrix $\mathbf{X}^T \mathbf{\Omega} \mathbf{X}$ can be inverted, and whether $\mathbf{S}$ can approximate a full-rank Jacobian matrix. This number is the ratio of highest to lowest singular value. It has the advantage of being a continuous function of the set of analogs, while the rank is a discontinuous function that takes only integer values. If the condition number is large, the set of analogs is almost contained in a plane, and the analogs might not be able to approximate the full Jacobian Φ . Conversely, if the condition number is close to 1, then the rank of the set of analogs is clearly 3, and analogs will be able to approximate the full Jacobian matrix. Note that the condition number of the set of analogs is not directly linked to the dimension of the attractor. One simply uses the fact that the attractor is locally close to a plane, without referring further to the complex notion of attractor dimension.

This can be investigated through numerical simulations of the L63 system, using a fourth-order Runge–Kutta numerical scheme and a time step of $\Delta t = 0.01$ to solve the governing equations. A catalog was generated from a trajectory of 10^5 nondimensional times, with the original equations ($\delta = 0$). LL analog forecasting was performed at horizon $t = 0.01$ with $K = 40$ analogs, on 10^4 points randomly selected on the attractor. The linear regression matrix $\mathbf{S}$ was then compared with $\mathbf{V}\Phi'$, with or without restriction to ($\mathbf{e}_1^a, \mathbf{e}_2^a$). To estimate numerically the real value of $\mathbf{V}\Phi'$, a third-order time expansion similar to Eq. (8) was computed directly from the model equations.

We use the same RMSE as before, where the value of N_{coeff} takes the restriction into account, such that it is either equal to 9 (full matrices) or 6 (restricted matrices).

Figure 5 shows that estimation of the Jacobian by the analogs improves as the catalog size (and therefore the catalog density) grows. This validates that the analogs are able to approximate accurately the Jacobian matrix of the flow map. The figure also shows that, once restricted to the two-dimensional subspace spanned by the analogs, this estimation is much more accurate and less fluctuating.

Figure 6 displays the RMSE of the full (3×3) matrix $\mathbf{S} - \mathbf{V}\Phi$ as a function of the condition number of the set of analogs. We can see in this figure that large RMSE values are highly correlated with high condition numbers, while low RMSE values can only be achieved when the condition number of the analogs is close to 1.

All these elements show that the estimation of the Jacobian matrix from analogs is highly dependent on the number of analogs K , the condition number of the set analogs, the attractor's dimension, and the phase-space dimension n . However, the fact

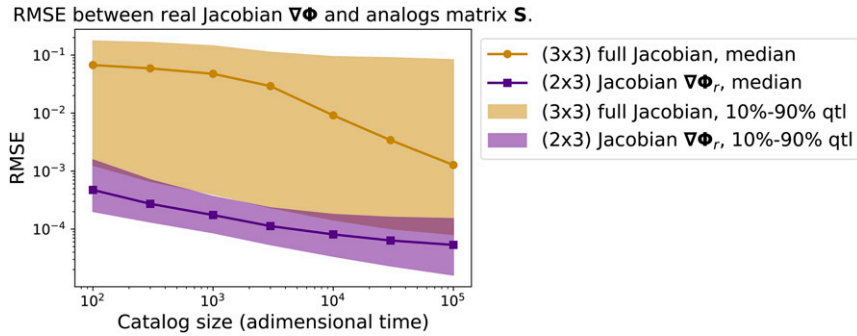


FIG. 5. RMSE in estimating with analogs the L63 Jacobian matrix, as a function of catalog size. The brown circles indicate the median RMSE (with 10% and 90% quantiles) of the total (3×3) Jacobian matrix. The violet squares indicate the median RMSE (with 10% and 90% quantiles) of the (2×2) Jacobian matrix after projection on the two first EOFs of the successors and restriction to the two first EOFs of the analogs. The projection restriction implies much lower RMSE, and a much lower variability. Both estimation errors are decreasing functions of the catalog size. The number of test points decreases with catalog size, as more test points are needed for small catalogs.

that the matrix from the LL operator does not approximate the full Jacobian $\nabla\Phi'$ does not mean that the analog forecast will poorly approximate the future state $\mathbf{x}_t$. For the LL forecast to be efficient, one only needs a good approximation of the restricted Jacobian, and that the inversion associated with the linear regression is not ill conditioned.

5) EFFECT OF OBSERVATIONAL NOISE

In the strong noise limit, variations of $\mathbf{a}_t^k$ (respectively $\mathbf{a}_0^k$) with k are dominated by variations of ϵ_t^k (respectively ϵ_0^k) and there is no correlation between successors and analogs. Therefore, the least squares estimate of the linear relationship $\mathbf{S}$ between the analogs and successors gives a matrix of zeros, and the LL operator reduces to the LC operator. However, this effect can be mitigated by a large value of K .

We conducted numerical experiments with the three-dimensional Lorenz system (not shown). We considered additive noises $\epsilon \sim \text{i.i.d. } \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, and σ is a given percentage (0.01 or 1) of the root-mean-squared distance between two points picked randomly on the attractor. We used the values $K = 40$ 200. In the strong noise case (1%), $\mathbf{S}$ is close to zero (for $K = 40$ and $K = 200$), and the condition number of analogs is always low because the two-dimensional structure of the attractor is lost inside the isotropic, three-dimensional noise. In the weak noise case (0.01%), we recover that $\text{RMSE}(\mathbf{S} - \nabla\Phi')$ is a growing function of the analogs' condition number as in Fig. 6. Also, $\text{RMSE}(\mathbf{S} - \nabla\Phi')$ is a decreasing function of K . However, even for low condition numbers and high K , the estimation error is much larger than in the noise-free case of Fig. 6.

c. Forecast error at various time horizons: Numerical experiment

In this section, we use Eqs. (9e), (10e), and (11a) to interpret analog forecast errors on a practical numerical example of the L63, for various lead times. In particular, we explore the limitations of these equations for large time horizons.

Before describing the numerical experiment, we recall some of the results of Farmer and Sidorowich (1988). The authors used analog-like forecasting techniques and focused on the long-term behavior of the forecast errors. In particular, they derived the time slope of the mean exponential growth of errors associated with the maximal Lyapunov exponent (MLE). As long as there is a nonzero component of $\mathbf{a}_0^k - \mathbf{x}_0$ along the direction associated with the MLE, this exponent drives the long-term amplitude of $\nabla\Phi' \cdot (\mathbf{a}_0^k - \mathbf{x}_0)$ and of the terms that involve the Hessian included in $\mathcal{O}(|\mathbf{a}_0^k - \mathbf{x}_0|^2)$ in Eqs. (9a),

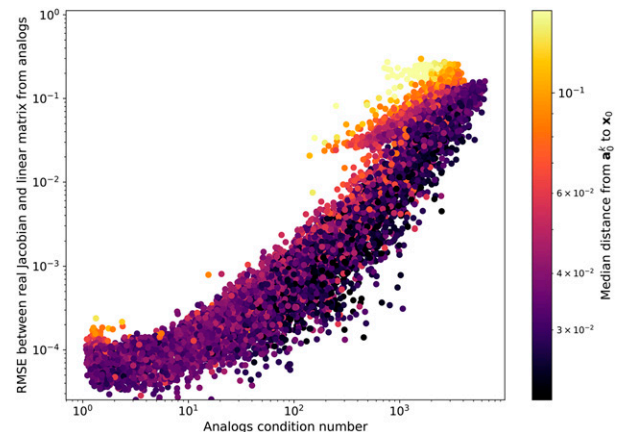


FIG. 6. RMSE in analogs estimation of the full (3×3) Jacobian matrix $\nabla\Phi'$ as a function of analog rank and median analog distance, with the L63 system. The rank of each set of analogs is measured by the ratio between the lowest and the highest singular value of the set of analogs. Most of the variability of the RMSE is explained by the rank of the analogs. Some of the remaining variability can be explained by the median distance from the analogs $\mathbf{a}_0^k$ to $\mathbf{x}_0$, which gives a measure of the local catalog density. The catalog size is 10^5 nondimensional times, $\delta = 0$, and we use $K = 40$ analogs. Tests are done at 10^4 points randomly selected on the attractor.

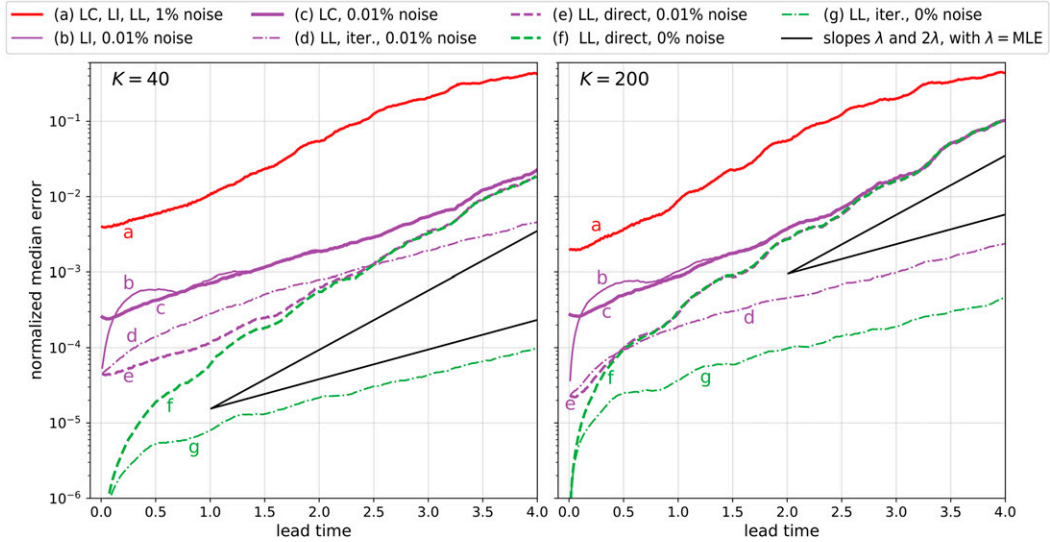


FIG. 7. Influence of noise intensity σ and number of analogs K on analog forecasting errors, using the Lorenz (1963) system, a catalog size of $L = 10^7$ and a time step $dt = 0.01$ between elements of the catalog. Median errors of mean analog forecasting operators $|\mu_{L_t} - \mathbf{x}_t|$ are computed over 2000 points taken randomly on the attractor. Black curves indicate slopes associated with the maximal Lyapunov exponent.

(10a), and (11a). At first order, the Jacobian grows as $e^{\lambda t}$, where λ is the MLE, and the Hessian grows as $e^{2\lambda t}$. Therefore, the errors of the LC and LI grow as $e^{\lambda t}$, and the error of the LL grows as $e^{2\lambda t}$. However, for very large lead times, the error term that grows as $e^{2\lambda t}$ becomes dominant and drives the error of all operators LC, LI and LL, such that they become equivalent in accuracy.

Farmer and Sidorowich (1988) also studied an iterative version of the LL operator. Instead of directly using successors at time t , one uses successors at time dt to split the forecast into multiple forecasts of small time step. This avoids having an error term involving the Hessian that grows as $e^{2\lambda t}$. In this case, at each iteration, the forecast error grows through both an additive error due to the nonzero difference $\mathbf{V}\Phi^{dt} - \mathbf{S}$, and the growth of previous errors through the dynamics of the system. For large lead times, the growth of previous errors becomes dominant, and grows with time as $e^{\lambda t}$.

We conducted numerical experiments of the L63 system to produce mean analog forecasts, whose errors were computed over 2000 points on the attractor. Additive, independent and identically distributed (i.i.d.) Gaussian observational noises were considered, and we set $\delta = 0$. Figure 7 shows medians of these forecast errors for several lead times and several methods.

First, let us consider the case of *strong noise* (1% of RMS distance in the attractor, red curves in Fig. 7). In this case and for the experiment reported here, the noise is larger than the typical distance between analogs and initial state $|\mathbf{a}_0^k - \mathbf{x}_0|$. All methods (LC, LI, and LL) give similar results (indistinguishable on the graph). The iterative LL method adds an error term at each iteration due to observational noise and is therefore less accurate than other methods (not shown). As stated above, in the case of strong noise $\mathbf{S} \approx 0$ and the LL operator tends to

the LC operator at all times. Then, for small lead times (here, $t \lesssim 1$), the noise terms of Eqs. (9e) and (10e) are dominant, such that the LC and LI operator have similar performances. This also explains why raising the value of K diminishes the forecast error. On the other hand, for large lead times (here, $t \gtrsim 1$), the terms proportional to $(\mu_0 - \mathbf{x}_0)$ in Eqs. (9e) and (10e) are dominant. Therefore, raising the value of K does not change much the accuracy of mean analog forecasting. Also, for large lead times, the growth of errors through the Jacobian $\mathbf{V}\Phi^t$ follows the slope given by the MLE (we take here the value of 0.9057; see Viswanath 1998). Finally, note that the similar performances of the LC and LI operators at large lead times can be explained by the fact that $|\mathbf{V}\Phi^t| \gg \|\mathbf{I}\|$.

Then, consider the case of *weak noise* (0.01% of RMS distance in the attractor, violet curves in Fig. 7). In this case, the noise amplitude is smaller than the typical distance between analogs and initial state, but still nonnegligible. For small lead times (here, $t \lesssim 1.5$), there is no dominant term in Eqs. (9e) and (10e). Thus, raising the value of K diminishes the errors due to noise but also raises the errors proportional to $(\mu_0 - \mathbf{x}_0)$, resulting in better performances for the LL, but similar performances for the LC and LI. For large lead times ($t \gtrsim 3.5$ for $K = 40$ and $t \gtrsim 2.5$ for $K = 200$), all methods LC, LI, and LL are equivalent and driven by the Hessian terms $\mathcal{O}(\sum \omega_k |\mathbf{a}_0^k - \mathbf{x}_0|^2)$ with the expected slope of 2λ . Therefore, raising the value of K increases bias and raises the mean analog forecasting errors. As expected, the iterative LL error grows as $e^{\lambda t}$ and is more accurate than the direct LL at large lead times ($t \gtrsim 2.5$ for $K = 40$ and 0.01% noise, $t \gtrsim 0.5$ for $K = 200$ and 0.01% noise, $t \gtrsim 0.1$ for $K = 40$ and zero noise, and at all times for $K = 200$ and zero noise). Note also that this iterative method is the only one that benefits from noise reduction through the increase of K at large lead times.

For very small lead times only ($t \leq 0.1$), the LI operator has better performances than the LC operator. Note that at very small lead times (also $t \leq 0.1$), the forecast error of the LC operator is a decreasing function of time. This is due to the fact that the system under study is dissipative and the trace of $\nabla \mathbf{f}$ is negative (for more details, see the work of Nicolis et al. 2009). Thus, at very small lead times the distance between two initially close states decreases. This decreasing trend at very small lead times is also witnessed for the direct LL operator. We interpret this fact as the result of a very good match between the real Jacobian and the linear matrix estimated by the direct LL. On the other hand, the iterative LL adds a new error term at each iteration and is thus not a decreasing function of time even for lead times $t \leq 0.1$. This explains why the iterative LL has poorer performances than the direct LL for these very small lead times. This can only be witnessed when noise is not the main driver of forecast error.

Similar arguments explain the curves of the LL operator and iterative LL operator for *zero noise* (green curves in Fig. 7). In this case, raising the value of K increases the error at all times. The fact that the curves of the direct LL with zero noise and with weak noise coincide at large lead times confirm that noise is not the main driver of forecast error at large lead times.

Note that these results are, in principle, influenced by the choice of the Euclidean metric to find analogs. Alternatives have been explored (e.g., Blanchet et al. 2018). But Toth (1991) argued that their performances marginally vary for forecast evaluations.

d. Evolution of mean and covariance under Gaussian assumption

In this section, it is assumed that the weighted discrete distributions of the analogs $\sum_k \omega_k \delta_{\mathbf{a}_0^k}$ and of their successors $\sum_k \omega_k \delta_{\mathbf{a}_t^k}$ can be approximated by Gaussian distributions:

$$\sum_k \omega_k \delta_{\mathbf{a}_0^k} \approx \mathcal{N}(\boldsymbol{\mu}_0, \mathbf{P}_0), \quad (16a)$$

$$\sum_k \omega_k \delta_{\mathbf{a}_t^k} \approx \mathcal{N}(\boldsymbol{\mu}_t, \mathbf{P}_t), \quad (16b)$$

where we have $\boldsymbol{\mu}_t = \boldsymbol{\mu}_{LC}$. One advantage of the Gaussian distribution is that any linear combination of Gaussian-distributed variables is also Gaussian distributed. This property allows us to derive many results in their exact, analytical form. This distribution also has the benefit of being represented entirely by its two first moments (i.e., its mean and variance). Finally, note that assuming Gaussian forecasts simplifies computations in a data-assimilation framework. However, this distribution can only represent light-tailed phenomena, and might thus not be suited to the study of extremes.

Note that if Eq. (16a) holds and if the flow map can be approximated by a linear function in the convex hull of the set of analogs, then Eq. (16b) holds. In practice, this is satisfied if the flow map is smooth enough, and if the catalog is dense enough so that most of the variations in the set of successors can be explained by the Jacobian matrix of the flow map at $\boldsymbol{\mu}_0$.

In this section, we assume zero observational noise. Nonzero observational, centered Gaussian noises would give additional

terms in the expressions given below for the covariance matrices. Combining the Gaussian hypothesis with Eq. (5b) and approximating $\Phi_a^t(\cdot)$ by its tangent around $\boldsymbol{\mu}_0$ we have the classic relationships:

$$\boldsymbol{\mu}_t = \Phi_a^t(\boldsymbol{\mu}_0) + \mathcal{O}(\text{Tr} \mathbf{P}_0), \quad (17a)$$

$$\mathbf{P}_t = \nabla \Phi_a^t|_{\boldsymbol{\mu}_0} \mathbf{P}_0 \nabla \Phi_a^t|_{\boldsymbol{\mu}_0}^T + \mathcal{O}(\text{Tr} \mathbf{P}_0), \quad (17b)$$

where Tr is the trace operator. Similar relations can be found using the differential representation of Eq. (5b):

$$\frac{d\boldsymbol{\mu}_t}{dt} = \mathbf{f}_a(\boldsymbol{\mu}_t) + \mathcal{O}(\text{Tr} \mathbf{P}_t), \quad \boldsymbol{\mu}_{t=0} = \boldsymbol{\mu}_0, \quad (18a)$$

$$\frac{d\mathbf{P}_t}{dt} = \nabla \mathbf{f}_a|_{\boldsymbol{\mu}_t} \mathbf{P}_t + \mathbf{P}_t \nabla \mathbf{f}_a|_{\boldsymbol{\mu}_t}^T + \mathcal{O}(\text{Tr} \mathbf{P}_t), \quad \mathbf{P}_{t=0} = \mathbf{P}_0. \quad (18b)$$

Now, let us make the simplifying hypothesis that $|\mathbf{x}_0 - \boldsymbol{\mu}_0|^2 \leq \text{Tr} \mathbf{P}_0$, which means that the state $\mathbf{x}_0$ is not farther from the analogs' mean $\boldsymbol{\mu}_0$ than the standard deviation of the analogs. Then, one evaluates Φ_a^t , $\mathbf{f}_a$ and their derivatives at $\mathbf{x}_0$ and $\mathbf{x}_t$ instead of $\boldsymbol{\mu}_0$ and $\boldsymbol{\mu}_t$, giving additional terms:

$$\boldsymbol{\mu}_t = \Phi^t(\mathbf{x}_0) + \delta \tilde{\Phi}^t(\mathbf{x}_0) + \nabla \Phi^t|_{\mathbf{x}_0} (\boldsymbol{\mu}_0 - \mathbf{x}_0) + \mathcal{O}(\text{Tr} \mathbf{P}_0, \delta |\boldsymbol{\mu}_0 - \mathbf{x}_0|), \quad (19a)$$

$$\begin{aligned} \mathbf{P}_t = & \nabla \Phi^t|_{\mathbf{x}_0} \mathbf{P}_0 \nabla \Phi^t|_{\mathbf{x}_0}^T + \delta (\nabla \tilde{\Phi}^t|_{\mathbf{x}_0} \mathbf{P}_0 \nabla \Phi^t|_{\mathbf{x}_0}^T \\ & + \nabla \Phi^t|_{\mathbf{x}_0} \mathbf{P}_0 \nabla \tilde{\Phi}^t|_{\mathbf{x}_0}^T) + ([(\boldsymbol{\mu}_0 - \mathbf{x}_0) \nabla^2 \Phi^t|_{\mathbf{x}_0}] \mathbf{P}_0 \nabla \Phi^t|_{\mathbf{x}_0}^T \\ & + \nabla \Phi^t|_{\mathbf{x}_0} \mathbf{P}_0 [(\boldsymbol{\mu}_0 - \mathbf{x}_0) \nabla^2 \Phi^t|_{\mathbf{x}_0}]^T) + \mathcal{O}(\text{Tr} \mathbf{P}_0, \delta |\boldsymbol{\mu}_0 - \mathbf{x}_0|), \end{aligned} \quad (19b)$$

where terms of order $|\boldsymbol{\mu}_0 - \mathbf{x}_0|^2$ are included in $\mathcal{O}(\text{Tr} \mathbf{P}_0)$ and $\nabla^2 \Phi^t|_{\mathbf{x}_0}$ is the Hessian of Φ^t at $\mathbf{x}_0$. In the time differential representation we have

$$\frac{d(\boldsymbol{\mu}_t - \mathbf{x}_t)}{dt} = \nabla \mathbf{f}|_{\mathbf{x}_t} (\boldsymbol{\mu}_t - \mathbf{x}_t) + \tilde{\mathbf{f}}(\mathbf{x}_t) + \mathcal{O}(\text{Tr} \mathbf{P}_t, \delta |\boldsymbol{\mu}_t - \mathbf{x}_t|), \quad (20a)$$

$$\begin{aligned} \frac{d\mathbf{P}_t}{dt} = & \nabla \mathbf{f}|_{\mathbf{x}_t} \mathbf{P}_t + \mathbf{P}_t \nabla \mathbf{f}|_{\mathbf{x}_t}^T + \delta (\nabla \tilde{\mathbf{f}}|_{\mathbf{x}_t} \mathbf{P}_t + \mathbf{P}_t \nabla \tilde{\mathbf{f}}|_{\mathbf{x}_t}^T) \\ & + ([(\boldsymbol{\mu}_t - \mathbf{x}_t) \nabla^2 \mathbf{f}|_{\mathbf{x}_t}] \mathbf{P}_t + \mathbf{P}_t [(\boldsymbol{\mu}_t - \mathbf{x}_t) \nabla^2 \mathbf{f}|_{\mathbf{x}_t}]^T) \\ & + \mathcal{O}(\text{Tr} \mathbf{P}_t, \delta |\boldsymbol{\mu}_t - \mathbf{x}_t|). \end{aligned} \quad (20b)$$

Equation (20a) is equivalent to Eq. (19a), which is also equivalent to Eq. (6b). Equation (20a) can be Taylor expanded around $t = 0$ to find Eq. (9b). This analysis recovers the results from section 4a for the mean forecast of the LC analog forecasting operator.

Equations (20b) and (19b) are two representations of the same phenomenon. They show that at first order, the growth in covariance between the analogs and successors is directly linked to the Jacobian matrix of Φ^t at $\mathbf{x}_0$. The covariance of the analog forecast will depend on the covariance of the analogs at $t = 0$, $\mathbf{P}_0$, and on the system's local Jacobian $\nabla \Phi^t|_{\mathbf{x}_0}$. This is another way to see that the analogs are highly linked to the local dynamics of the system. If the local dynamics induce a large spread in the future possible trajectories, it is captured in

the successors' covariance $\mathbf{P}_t$. On the contrary, if the local dynamics are flat ($\nabla\Phi'|_{\mathbf{x}_0} \simeq \mathbf{I}$ or $\nabla\mathbf{f}|_{\mathbf{x}_0} \simeq \mathbf{0}$) the successors' covariance is equal to the analogs' covariance.

Finally, in the presence of observational noise as in Eq. (5d) and if the latter are centered, Gaussian, of the form $\epsilon_0^k \sim \text{i.i.d.}\mathcal{N}(\mathbf{0}, \Sigma_0)$, $\epsilon_t^k \sim \text{i.i.d.}\mathcal{N}(\mathbf{0}, \Sigma_t)$, and independent of the analogs and successors, then Eq. (17b) is modified to give

$$\mathbf{P}_t = \nabla\Phi'_a|_{\mu_0}(\mathbf{P}_0 + \Sigma_0)\nabla\Phi'_a|_{\mu_0}^T + \Sigma_t + \mathcal{O}(\text{Tr}\mathbf{P}_0), \quad (21a)$$

while Eq. (17a) is unchanged. Similarly, Eq. (19b) is modified to give

$$\begin{aligned} \mathbf{P}_t = & \nabla\Phi'|_{\mathbf{x}_0}(\mathbf{P}_0 + \Sigma_0)\nabla\Phi'|_{\mathbf{x}_0}^T + \delta(\nabla\tilde{\Phi}'|_{\mathbf{x}_0}(\mathbf{P}_0 + \Sigma_0)\nabla\Phi'|_{\mathbf{x}_0}^T \\ & + \nabla\Phi'|_{\mathbf{x}_0}(\mathbf{P}_0 + \Sigma_0)\nabla\tilde{\Phi}'|_{\mathbf{x}_0}^T) + [(\mu_0 - \mathbf{x}_0)\nabla^2\Phi'|_{\mathbf{x}_0}] \\ & \times (\mathbf{P}_0 + \Sigma_0)\nabla\Phi'|_{\mathbf{x}_0}^T + \nabla\Phi'|_{\mathbf{x}_0}(\mathbf{P}_0 + \Sigma_0)[(\mu_0 - \mathbf{x}_0)\nabla^2\Phi'|_{\mathbf{x}_0}]^T \\ & + \Sigma_t + \mathcal{O}(\text{Tr}\mathbf{P}_0, \delta|\mu_0 - \mathbf{x}_0|). \end{aligned} \quad (21b)$$

These equations show that observational noise adds variability among the set of successors both through the observational error of the successors and the observational error of the analogs. Again, the noise contribution from the analogs to the successor variability is dependent on the local dynamics of the system.

5. Conclusions

For practical applications, one has often not access to analogs in phase space (i.e., full observation of the system). The analysis of partially observed systems corresponds to building (possibly) multivariate observables, to create a one-to-one map between the space of observations and the phase space. Such an analysis is beyond the scope of this article, and we assume that this issue has been treated separately. The dimension of observables is connected to the dimension of the full system, as demonstrated by [Caby et al. \(2020\)](#). The number of key variables in phase space has been deemed to be moderate (5–17; see [Faranda et al. 2017](#)). This motivates our investigation on low dimensional systems, in order to achieve meaningful illustrations. For theoretical works on analog forecasting with partial observations, the reader is referred to the early works of [Farmer and Sidorowich \(1988\)](#), [Sauer \(1994\)](#), and [Sugihara \(1994\)](#) that make use of time embeddings [see the study of [Sauer et al. \(1991\)](#) on time embeddings], and the more recent works of [Zhao and Giannakis \(2016\)](#) and [Alexander and Giannakis \(2020\)](#) that use kernel projection techniques. A complementary concern is the compensation of observational noise with analogs, which was investigated by [Chau et al. \(2020\)](#) in a data assimilation framework.

One must bear in mind that the use of analog forecasting in applications implies issues such as the choice of the space in which forecasting is performed, the choice of the right metric to compare analogs and initial state, and the combination of analogs with other techniques. In data assimilation, one might want to use Gaussian distributions instead of the discrete distributions of [section 2b](#) in order to use Kalman filtering.

Ridge and least absolute shrinkage and selection operator (LASSO) regularizations could be used instead of the techniques mentioned in [section 4b](#). Such regularization techniques use additional terms in the resolution of the least squares problem ([Hastie et al. 2009](#)) that prevent some complications of the linear regression, especially when the predictors are highly correlated. Each of these operational choices must be made accounting for memory use and computational time [see [Lguensat et al. \(2017\)](#) for differences between regular and coordinate-by-coordinate analog forecasting].

Analog forecasting allows us to avoid solving complex nonlinear equations by using existing solutions starting from similar initial conditions. The accuracy of analog forecasting depends on local dynamical properties of the system of interest. In particular, the quality of analog forecasts is related to the Jacobian matrix of the real system's flow map, and the linear regression from analogs to successors is shown to provide an approximation of this matrix. This approximation helps to study the mean accuracy of known analog forecasting operators, and to compare different methods that evaluate this Jacobian matrix, using numerical experiments of well-known dynamical systems. The LL operator is found to give the best approximation of the future state, provided that the linear regression is not ill posed. For long-term forecasts, an iterative version of the LL can be even more accurate. The LI operator is shown to have better skills than the LC at small lead times. The behavior of analog forecast errors at large lead times is driven by the maximal Lyapunov exponent of the system. Additive observational noise lowers the performances of analog forecasts but can be mitigated by using a larger number of analogs. However, raising the number of analogs also increases bias, and the optimal trade-off depends on the time horizon and the analog method used. The Jacobian matrix of the flow map is found to drive the growth of the successors' covariance matrix. Altogether, this brings theoretical evidence that analogs can be used to emulate a real system and gives quantitative expressions to analyze and predict the accuracy of analog forecasting techniques.

Acknowledgments. The work was financially supported by ERC Grant 338965-A2C2 and ANR 10-IEED-0006-26 (CARAVELE project). We thank the three anonymous reviewers, as this version of the paper owes much to their remarks and suggestions.

APPENDIX A

Lorenz Systems

The three-variable L63 ([Lorenz 1963](#)) system of equations is

$$\begin{cases} \frac{dx_1}{dt} = \sigma(x_2 - x_1), \\ \frac{dx_2}{dt} = x_1(\rho - x_3) - x_2, \\ \frac{dx_3}{dt} = x_1x_2 - \beta x_3, \end{cases} \quad (A1)$$

with usual parameters $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$.

The n -variable L96 (Lorenz 1996) system of equations is

$$\forall i \in [1, n], \quad \frac{dx_i}{dt} = -(x_{i-2} + x_{i+1})x_{i-1} - x_i + \theta, \quad (\text{A2})$$

where θ is the forcing parameter. We set $n = 8$ and $\theta = 8$, and use periodic boundary conditions $x_{i+n} = x_i$.

APPENDIX B

Product of Hessian with Vectors

Let $\mathbf{g}$ be a vector-valued, phase-space-dependent function $\mathbf{g}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such as Φ^t or $\mathbf{f}$.

The Hessian of $\mathbf{g}$ at $\mathbf{x}$ is noted $\nabla^2 \mathbf{g}|_{\mathbf{x}}$. It is of dimension n^3 and its (i, j, k) th coefficient $[\nabla^2 \mathbf{g}|_{\mathbf{x}}]_{i,j,k}$ equals $[\partial^2 g_k / (\partial x_i \partial x_j)](\mathbf{x})$. The product of a Hessian $\nabla^2 \mathbf{g}|_{\mathbf{x}}$ with a n -dimensional vector $\mathbf{y}$ is a matrix and its (i, k) th coefficient $[\mathbf{y} \nabla^2 \mathbf{g}|_{\mathbf{x}}]_{i,k}$ equals $\sum_j y_j \partial^2 g_k / (\partial x_i \partial x_j)$. The double product of a Hessian with two n -dimensional vectors $\mathbf{y}$ and $\mathbf{z}$ is a vector and its k th coefficient $[\mathbf{y}(\nabla^2 \mathbf{g}|_{\mathbf{x}})\mathbf{z}]_k$ equals $\sum_{i,j} y_i z_j \partial^2 g_k / (\partial x_i \partial x_j)$. The double product of a Hessian $\nabla^2 \mathbf{g}|_{\mathbf{x}}$ with two matrices $\mathbf{X}$ and $\mathbf{Y}$ of same shape $k \times n$ is a matrix of shape $k \times n$, and its (k, j) th coefficient is $\sum_{l,m} X_{k,l} Y_{l,m} \partial^2 g_j / (\partial x_l \partial x_m)$.

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